

ITT320 INTRO TO COMPUTER SECURITY

Chapter 9 : Defending Against Trojan Horses, Adware & Spyware

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Objectives

- Understanding Trojan horses
- How a Trojan Horses Works
- Example of Trojan Horses : Old vs New
- Take steps to prevent Trojan horse attacks
- Describe spyware
- Use anti-spyware software
- Create anti-spyware policies



Introduction

Though not as common as viruses, Trojan horses still pose a real threat to computer systems.

Spyware and adware continue to grow and clutter computer networks and individual computers. This chapter provides ways to combat these particular types of threats.

Understanding Trojan Horses

A Trojan horse or Trojan is a type of malware that is often disguised as legitimate software. A program that looks benign, but is not.

Typical actions Trojan horses take:

- Delete files from a computer | turn on camera
- Download harmful software | Zombie
- Install a key logger / Key strokes
- Spread other malware / install Ransomware
- Use the computer to launch a DDoS
- Search for personal information
- Install “back door” to the computer



How a Trojan horse works

In order to get infected, user must download the malicious application and run/install it.

Typical modus of Trojan horses:

- Social Engineering - used to convince end users to download the malicious application
- Banner ads, Website Links Pop-Up Ads
- Email Spamming – with Attachment files that contains the trojan
- BotNet- continue spreading the Trojan horse through Zombies



Identifying Trojan Horses

Example of some of the famous trojan horses that have been attacked us around the world.

- ✓ Back Orifice
- ✓ Internet Explorer Trojan Horse
- ✓ NetBus
- ✓ Linux Trojan Horses
- ✓ Portal of Doom

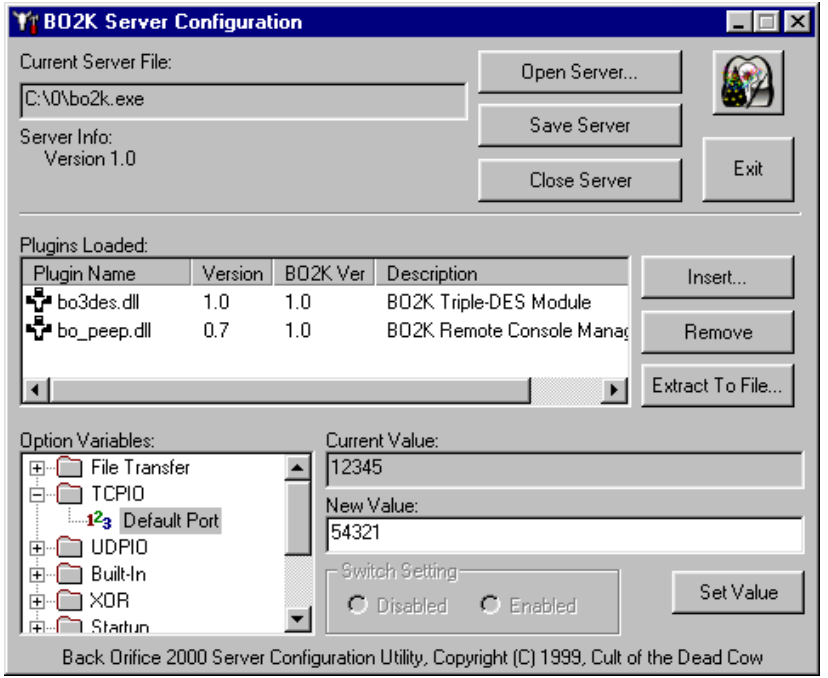


Back Orifice

Back Orifice is a computer program designed for remote system administration. It enables a user to control a computer running the Microsoft Windows operating system from a remote location.

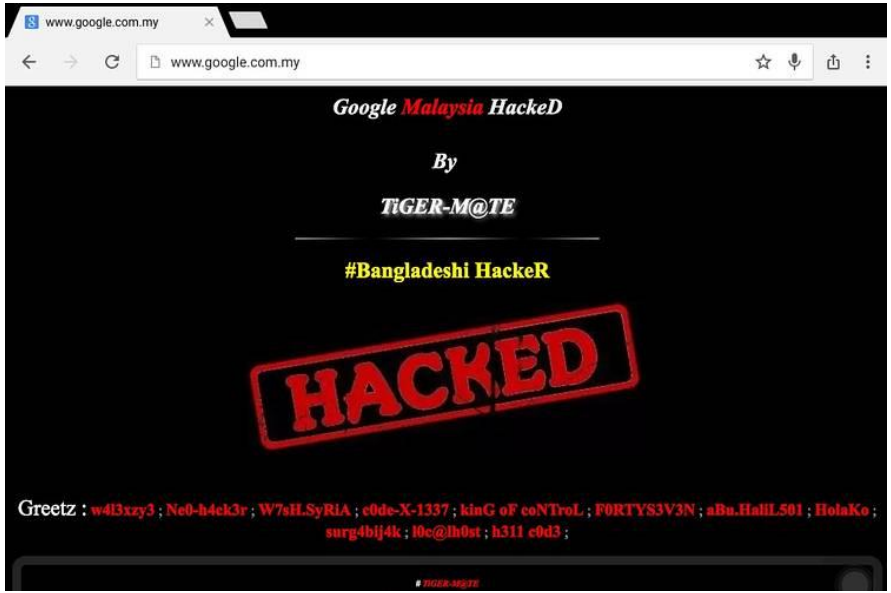
It Has been misused to embed in trojan and create connection to the victim's computer.

- Allows control over TCP/IP
- Entirely self-installing
- Can be attached to legitimate applications
- Does not appear in the task list
- Registry is the best way to remove



Internet Explorer Trojan Horse

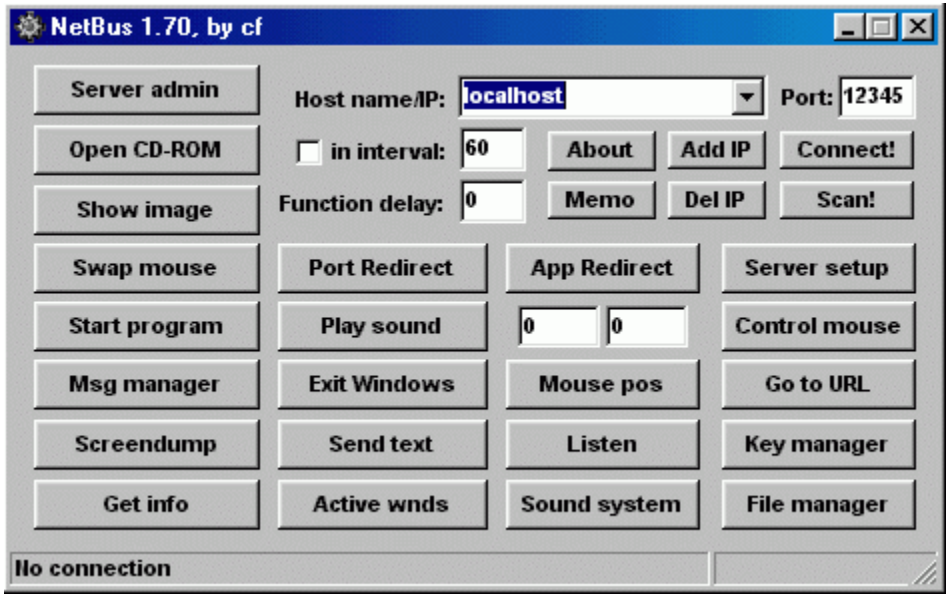
- Released in 2003
- Targets Microsoft's Internet Explorer Browser
- Changes the DNS configuration on the Windows machine
- Redirects requests to the hacker's site
- Patch released by Microsoft
- Check out Secunia-OSI to see if your browser is vulnerable



NetBus

NetBus or **Netbus** is a software program for remotely controlling a Microsoft Windows computer system over a network.

- Similar to Back Orifice
- Only works on port 20034
- Simple to check infection
- Removal through the Registry
- Easy-to-use GUI



Portal of Doom

Hijacks the computers of unsuspecting Windows users running old operating systems. Computers running Windows 95, 98, ME, NT, XP and Vista are vulnerable to the **Trojan**.

Port 9872 - 9875

Back door tool allows remote users to perform the following:

- ✓ Open and close the CD tray
- ✓ Shut down the system
- ✓ Open files or programs
- ✓ Access drives
- ✓ Change passwords
- ✓ Log keystrokes
- ✓ Take screen shots

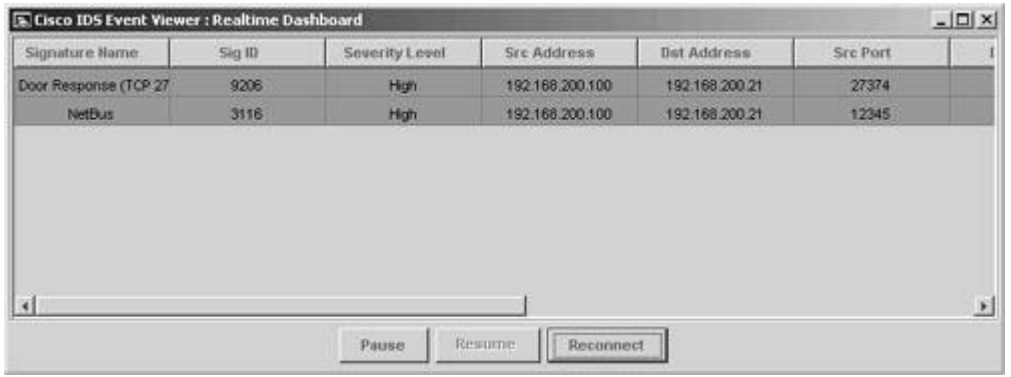


Image By : https://flylib.com

Linux Trojan Horses

- One released in 1999
- Typical back door Trojan
- Uploaded to at least one FTP server
- Not known how many systems were compromised

Other example : Twart

```
if ( a1 )
{
  v10 = strlen(s) + 277;
  v2 = alloca(16 * ((v10 + 23LL) / 0x10uLL));
  dest = (16 * ((&s + 7) >> 4));
  if ( dest )
  {
    strcpy(dest, s);
    strcat(dest, ".[redacted]");
    v3 = rand();
    sprintf(src, "-%08x", v3);
    strcat(dest, src);
    if ( fsize(dest) == -1 )
    {
      stream = fopen64(s, "r+");
      if ( stream )
      {
        v9 = fsize(s);
        if ( v9 )
        {
          if ( v9 > 15 )
          {
            mbedtls_aes_init(aes_ctx);
            pthread_mutex_lock(&csPreData);
            qmemcpy(ptr, &g_RansomHeader, 0x200uLL);
            mbedtls_aes_setkey_enc(aes_ctx, &g_KeyAES, 256LL);
            pthread_mutex_unlock(&csPreData);
            if ( !fseek(stream, 0LL, SEEK_END)
              && fwrite(ptr, 1uLL, 0x200uLL, stream)
              && !fseek(stream, -512 - v9, 1)
              && ProcessFileHandleWithLogic(stream, aes_ctx, a2, v9, CryptOneBlock) )
            {
              v13 = 1;
            }
```

RansomEXX Trojan attacks Linux systems

Example of Current Trojan Horses

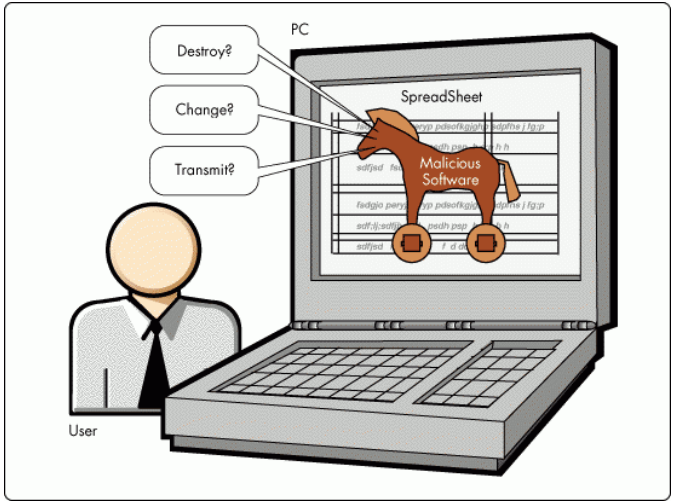
Some of the most famous discoveries include in the last 5 years :

- ✓ Bitfrost, a remote access Trojan (RAT) – Windows, altering components
- ✓ Tiny Banker - steal sensitive financial information
- ✓ FakeAV Trojan – when users followed directions to fix the problem by FakeAV, then they actually downloaded more malware
- ✓ Magic Lantern – Created by FBI, government trojan to record keystrokes logging
- ✓ Zues - a financial services crimeware toolkit allows attacker to build their own trojan horses

https://threats.kaspersky.com/en/threat/

Symptoms of a Trojan Horse

- Home page for your browser changes
- Any change to passwords, usernames, accounts, etc.
- Any change to screen savers
- Changes to mouse settings, backgrounds, etc.
- Any device seeming to work on its own





Preventing Trojan Horses

The answer is a hybrid approach using:

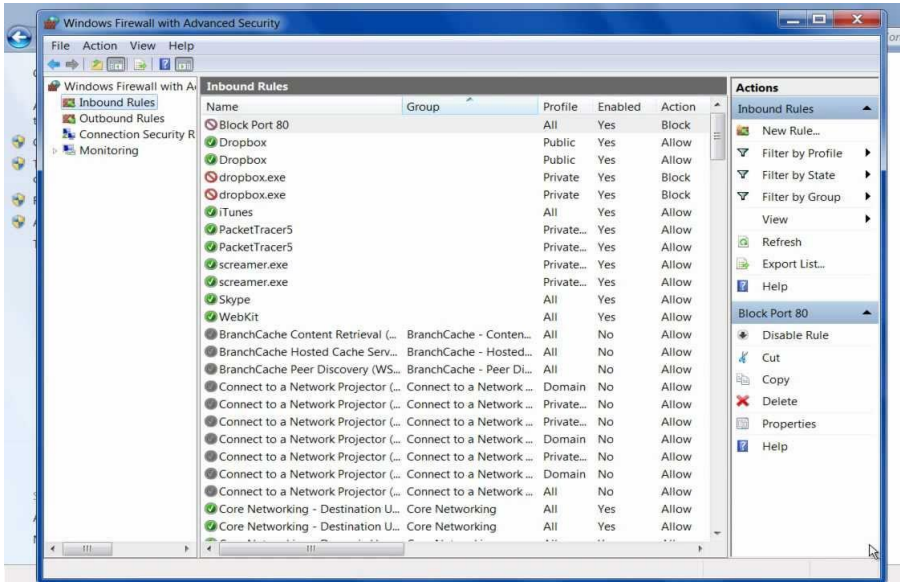
1. Technological measures
2. Policy measures

Technological Measures

1. Block unneeded ports (e.g. 20034)
2. Utilize anti-virus software (most check for Trojans)
3. Prevent active code in browsers
4. Limit user's rights to just what is needed
5. Firewall
6. Patch Management

Blocking ports using Windows Firewall:

1. Tap on the Windows-key, type Windows Firewall, and select Windows Firewall with Advanced Security from the results.
2. Click on Inbound Rules when the firewall window opens.
3. Select New Rule from the Actions pane.
4. Select **Port** from the Rule Type listing.





Trojan Horse and Associated Port(s)

Table 10.1 Ports used by well known Trojan Horses	
Port(s) Used	**Trojan Horse**
57341	NetRaider
54320	Back Orifice 2000
37651	Yet Another Trojan (YAT)
33270	Trinity
31337 and 31338	Back Orifice
12624	Buttman
9872-9872, 3700	Portal of Doom (POD)
7300-7308	Net Monitor
2583	WinCrash

Policy Measures

- Never download any (email) attachments unless absolutely certain they are safe or expected
- If a port is not needed, close it
- Restrict the downloading of software
- Be cautious of hidden file extensions



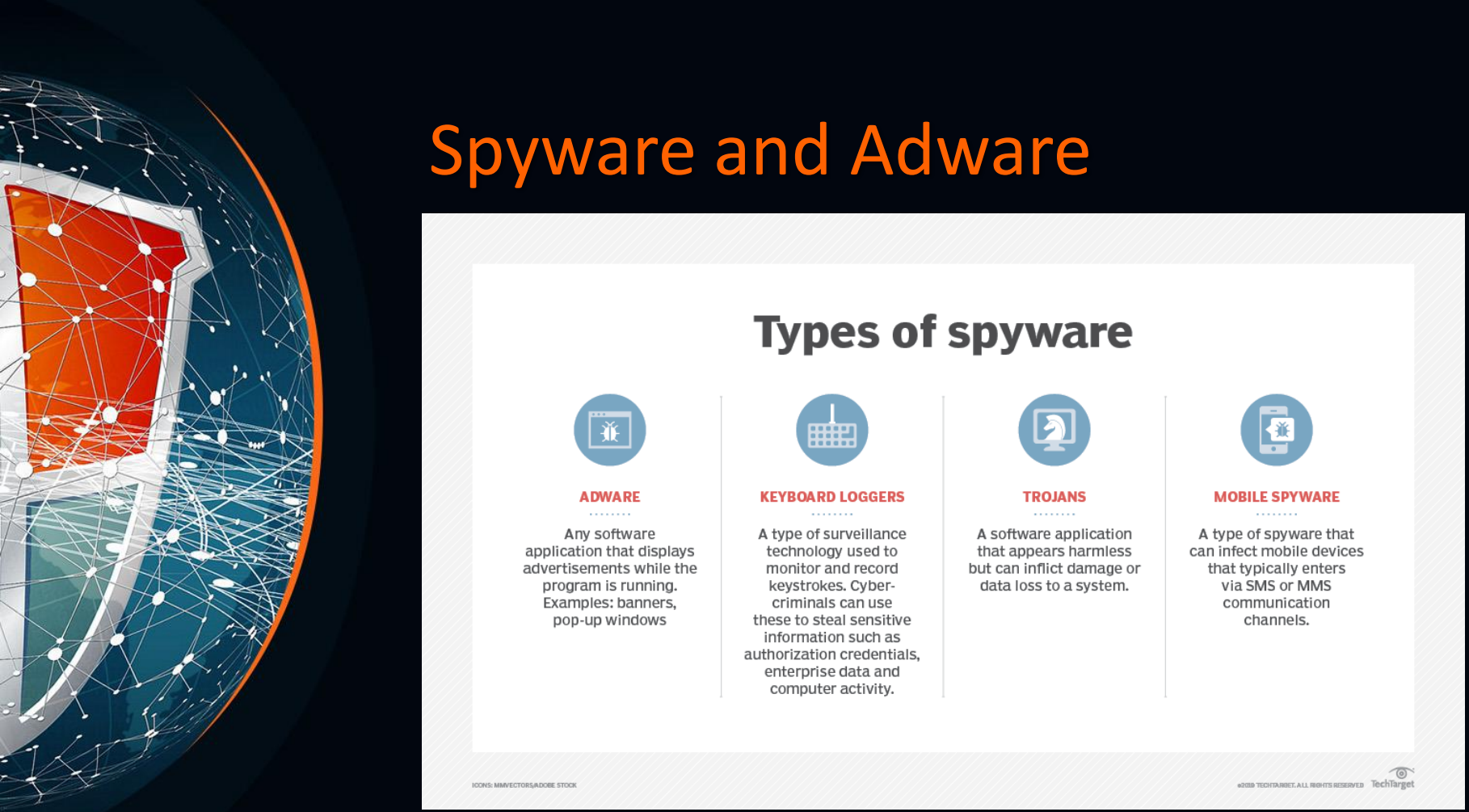


Spyware and Adware

Spyware is Part of Malware – installed without user authorization.

- Becoming more and more intrusive
- Can cause systems to crash
- Made to gather information and send it to third-parties
- Generate Pop-Ups not detected by pop-up blockers
- Monitors internet activities
- Track login credentials
- Spies sensitive information / credit card

Spyware and Adware





Identifying Spyware and Adware Example;

Like viruses and Trojan horses, spyware and adware programs become well known too ..

Gator (Adware)

- Two methods of removal

✓ Add/Remove Programs

✓ The Registry

RedSheriff (Spyware)

- Twofold problem:

✓ No one is certain what data is collected (except Manufacturer)

✓ Many people have a negative reaction to web site monitoring



Anti-Spyware

Old

- ✓ Spy Sweeper (www.webroot.com)
- ✓ Spyware Doctor (www.pctools.com/spywaredoctor/)
- ✓ Zero Spyware
- ✓ Microsoft Anti-Spyware (www.microsoft.com/athome/security/spywar e/software/default.mspx)

New

- ✓ **Malwarebytes** - scan through registry files, running programs, hard drives and individual files
- ✓ **Trend Micro HouseCal -** uses minimal processor and memory resources
- ✓ **Windows Defender -** lightweight antimalware tool that protects against threats such as spyware, adware and viruses

Anti-Spyware Policies

1. Never download any attachments you are not certain is safe
2. Only downloading software from trusted sources
3. Configure browser to block cookies
4. Configure browser to block scripts
5. Utilize browser pop-up blockers
6. Reading all disclosures when installing software
7. Avoiding interaction with pop-up ads; and
8. Staying current with updates and patches for browser, OS and application software



Anti-Spyware Policies Cont

Never download the following if you are uncertain of their safety:

- Applications
- Browser skins
- Screen savers
- Utilities



Block Java applets, or require manual approval of such



SUMMARY

- Both Trojan horses and spyware pose significant dangers
- Virus scanners and appropriate policies are your only protection against Trojan horses and spyware
- Carefully develop and implement anti-Trojan horse policies
- Spyware and Adware are growing problems for networks
- Spyware can compromise security
- Confidential information can be compromised by spyware
- Adware is more a nuisance than a real security threat:
  - However, there is a threshold of adware that can make a system unusable



SUMMARY

- There are numerous utilities that can help protect against Trojan horses (Anti-virus software)
- Available utilities can protect against spyware and adware
- Policies can work in conjunction with utilities to further protect systems